CATHOLIC JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

Higher 1

ANSWER SCHEME

CANDIDATE NAME							
CLASS	2T	INDEX NUMBER					

BIOLOGY

8876/01

Paper 1 Multiple Choice

17 SEPTEMBER 2021

1 hour

Additional Materials: OMR Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write and/or shade your name, NRIC / FIN number and HT group on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft 2B pencil** on the separate **OMR Answer Sheet.**

Read the instructions on the OMR Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

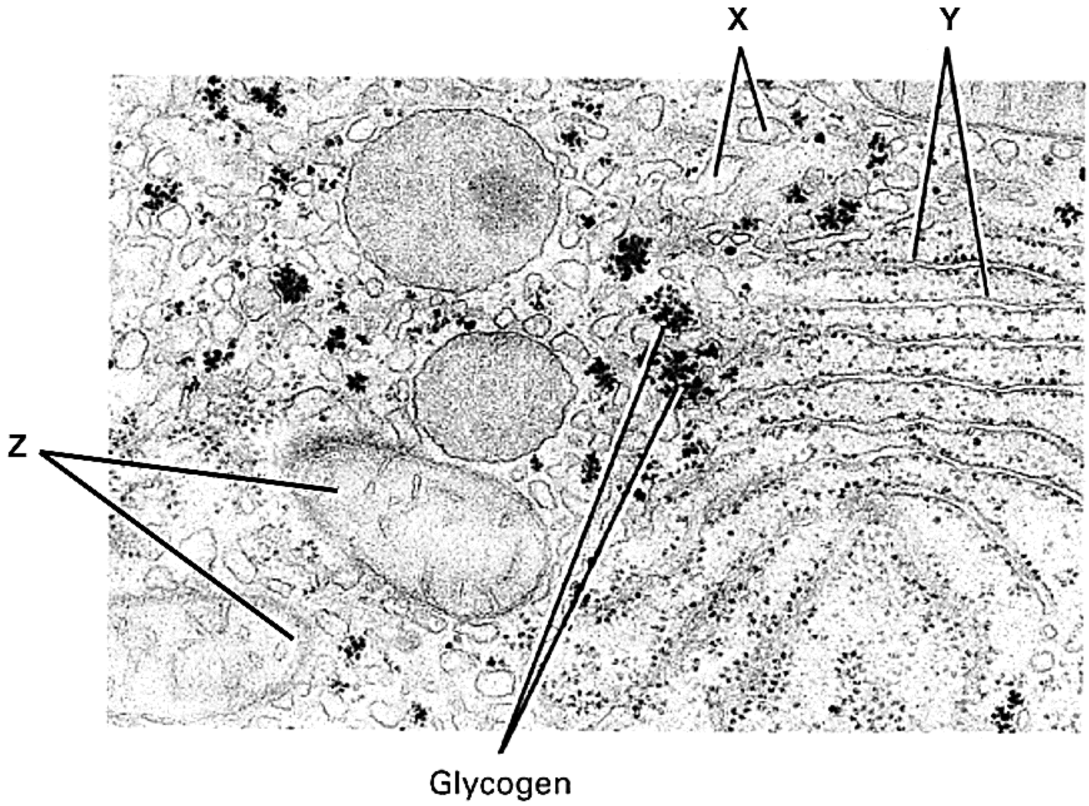
The use of an approved scientific calculator is expected, where appropriate.

1	D	6	C	11	A	16	D	21	C	26	D
2	C	7	A	12	D	17	C	22	D	27	C
3	C	8	B	13	B	18	D	23	D	28	A
4	D	9	B	14	B	19	D	24	D	29	B
5	A	10	A	15	A	20	A	25	C	30	D

This document consists of **18** printed pages and **0** blank pages.

Answer **all** questions

1 The figure below shows an electron micrograph of a cell which synthesizes glycogen. Glycogen is a form of energy storage polysaccharide in animal cells.

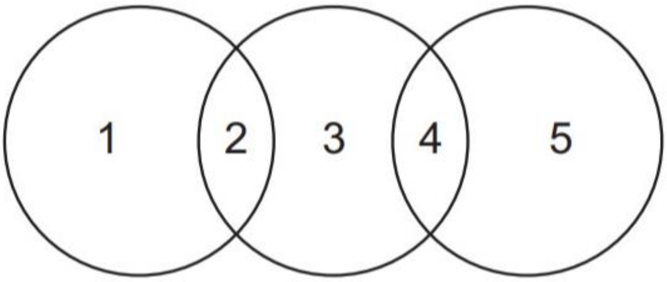


Which of the following options show the correct functions of structures **X, Y** and **Z**?

	X	Y	Z
A	Transporting of proteins for modification	Allow for modification, sorting and packaging of vesicles	Produce ATP for cellular activities
B	Digestion of worn out organelles	Site of protein synthesis	Produce ATP for Calvin cycle
C	Transporting of proteins out of the cell	Allow for the formation of secretory vesicles	Allow for light dependent reaction to occur
D	Detoxification of drugs	Packaging of proteins into transport vesicles	Produce ATP for cellular activities

ANS: D

2 The diagram shows some similarities between chloroplasts, mitochondria and typical prokaryotes.



Which row is correct?

	1	2	3	4	5
A	chloroplasts	circular DNA	mitochondria	80S ribosomes	prokaryotes
B	chloroplasts	80S ribosomes	mitochondria	circular DNA	prokaryotes
C	prokaryotes	circular DNA	mitochondria	circular DNA	chloroplasts
D	prokaryotes	70S ribosomes	chloroplasts	80S ribosomes	mitochondria

ANS: C

3 The table lists the features of two biological molecules found in eukaryotic cells.

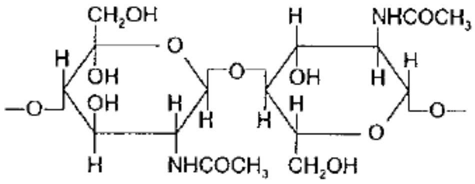
Molecule	Exists as a polymer	Hydrogen bonds are formed within the molecule	Remains in the cellular compartment where it is synthesised
X	Yes	Yes	Yes
Y	No	No	No

Which of the following shows the possible identities of molecules **X** and **Y**?

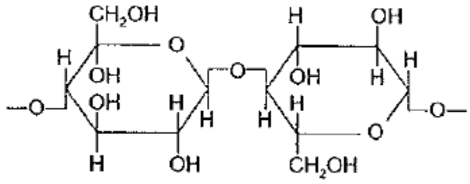
	Molecule **X**	Molecule *Y*
A	Triglyceride	Cellulose
B	Deoxyribonucleic acid	Collagen
C	Haemoglobin	Phospholipid
D	RNA polymerase	Cholesterol

ANS: C

4 The diagrams show short sections of some common polysaccharides and modified polysaccharides.



1



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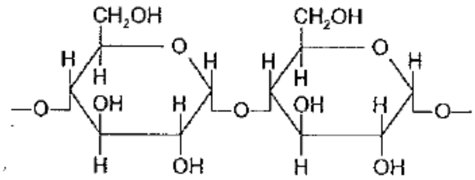

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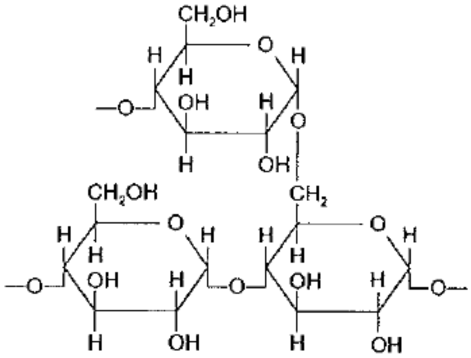

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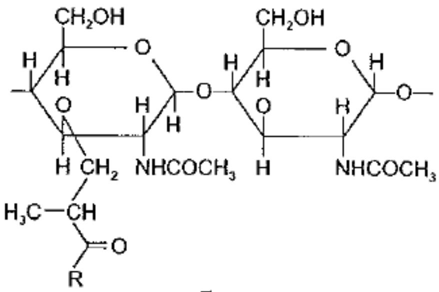


5

The polysaccharides can be described as below:

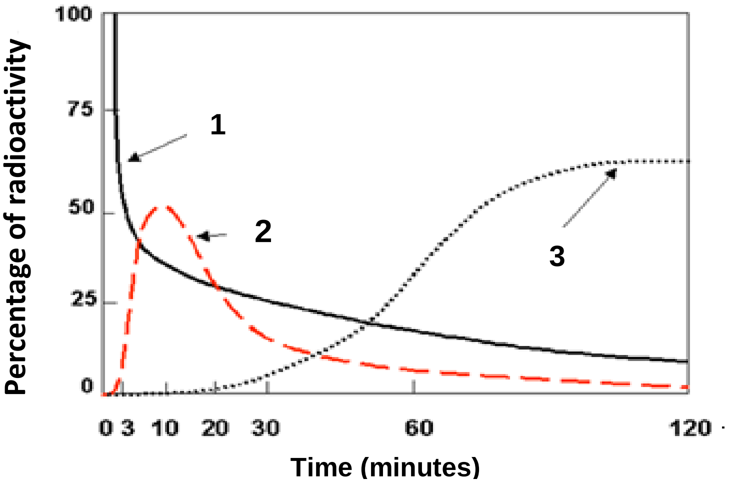
Which shows the correct pairings of the polysaccharide names and diagrams?

	Cellulose	Amylopectin	Peptidoglycan	Amylose	Chitin
A	**1**	**5**	**4**	**2**	**3**
B	**2**	**4**	**1**	**3**	**5**
C	**3**	**5**	**2**	**4**	**1**
D	**2**	**4**	**5**	**3**	**1**

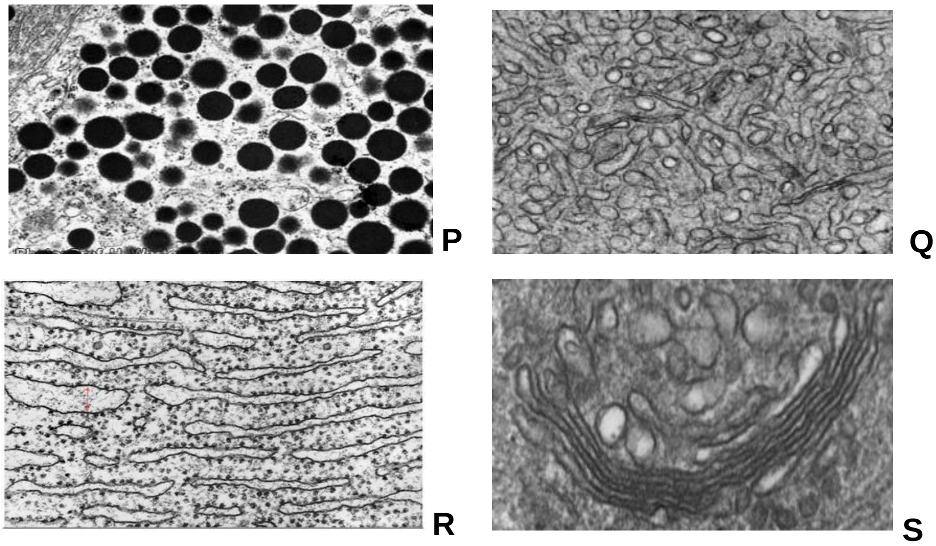
ANS: D

5 A pulse-chase experiment was used to trace the path taken by amino acids in a cell over a period of time. During the experiment, a cell producing salivary amylase was grown in a culture containing radioactive amino acids for a few minutes.

The percentage of radioactivity at the various organelles was then measured and the results of the experiment are shown in the graph below.



The micrographs of four different organelles from the cell producing salivary amylase are shown below.



Each graph corresponds to an organelle in the cell where the radioactivity was measured.

Which of the organelles shown above are correctly matched with the graphs?

	Graph **1**	Graph **2**	Graph **3**
A	**R**	**S**	**P**
B	**P**	**R**	**S**
C	**Q**	**R**	**S**
D	**R**	**S**	**Q**

ANS: D

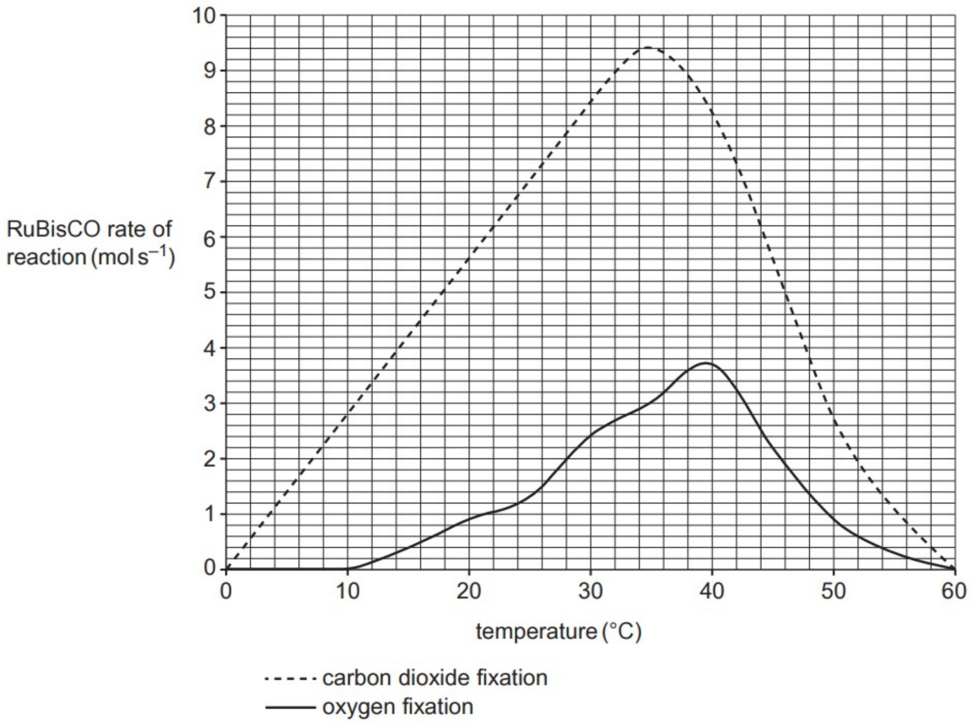
The following information is used for both questions **6** and **7**.

Ribulose bisphosphate carboxylase-oxygenase (RuBisCO) is an enzyme involved photosynthesis. In the process of carbon fixation, RuBisCO incorporates CO_2 into ribulose bisphosphate (RuBP) to form a 6-carbon compound.

Plants carry out gaseous exchange with the environment via their stomata. When the stomata is open, CO_2 diffuses in while O_2 and water vapour diffuses out. Under hot and dry conditions, plants close their stomata to reduce water loss.

When O_2 concentration is high and CO_2 concentration is low, RuBisCO binds to O_2 instead of CO_2.

6 The graph below shows how the carbon dioxide and oxygen fixing activities of RuBisCO are affected by temperature.



What are the correct percentage changes in RuBisCO carbon dioxide and oxygen fixing activities between 30°C and 40°C?

A carbon dioxide fixation -12.7%, oxygen fixation 23.3%

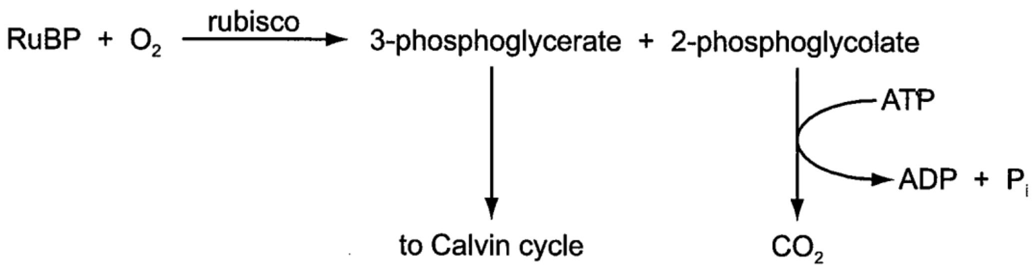
B carbon dioxide fixation -14.6%, oxygen fixation 18.9%

C carbon dioxide fixation -2.4%, oxygen fixation 54.2%

D carbon dioxide fixation -3.6%, oxygen fixation 35.1%

ANS: C

7 When O_2 binds to RuBisCO at its active site, this initiates the process known as photorespiration as shown below.



Which statement is supported by these facts?

A When O_2:CO_2 ratio is high, O_2 outcompetes CO_2 for the RuBisCO active site, resulting in photorespiration.

B Photorespiration is beneficial to plants as Calvin Cycle can proceed as before.

C CO_2 has a higher affinity than O_2 for the RuBisCO active site, allowing carbon fixation to take place most of the time.

D Photorespiration serves to reduce water loss for plants under hot and dry conditions.

ANS: A

8 Influenza virus has an enzyme called neuraminidase which breaks down glycoproteins in the membrane of the cell that the virus will infect. The glycoprotein binds to the active site of neuraminidase by induced fit.

Which statements about the induced fit hypothesis of enzyme action are correct?

1 The active site must have a complementary shape to the substrate for them to bind together.

2 Activation energy is lowered when enzyme-substrate complex is formed, just like in the lock and key mechanism.

3 The substrate is converted to product by specific R-groups in the active site just like the lock and key mechanism.

A **1** and **2**

B **2** and **3**

C **2** only

D **3** only

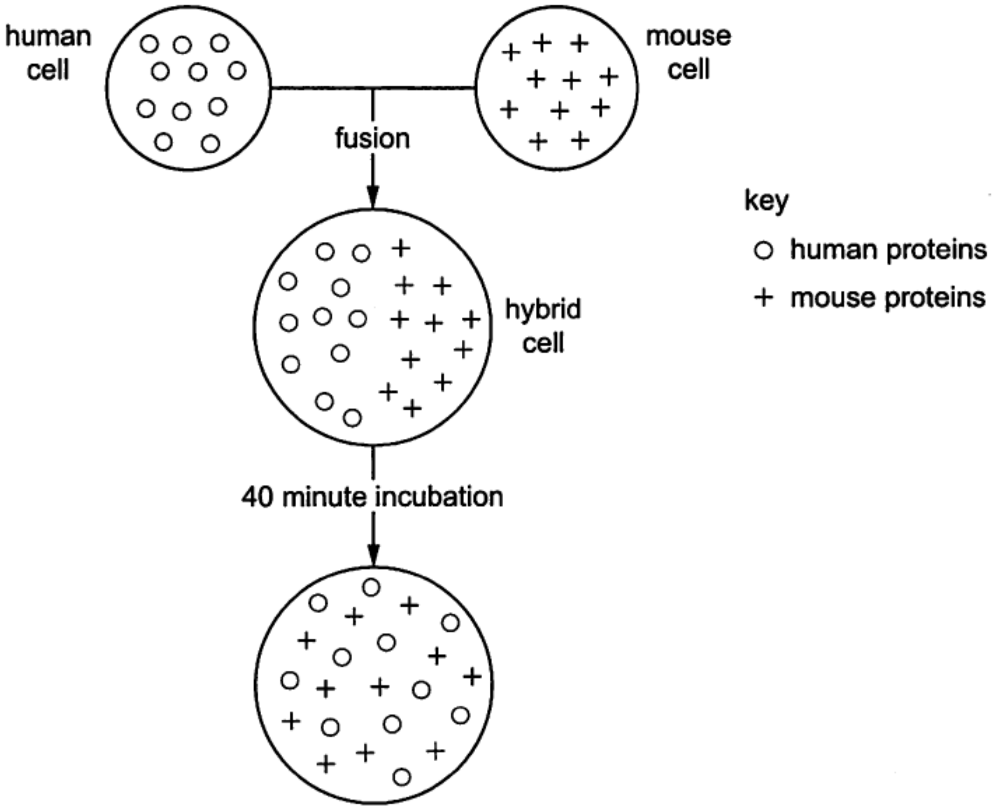
ANS: B

9 In an experiment, fluorescent dyes were attached to proteins on the outer surface of cell surface membranes. Fluorescent dyes of one colour were attached to proteins of a living human cell and fluorescent dyes of a different colour were attached to proteins of a living mouse cell.

The human cell and the mouse cell were then fused to form a hybrid cell.

At first, the proteins attached to different fluorescent dyes remained separate, but after 40 minutes the proteins were distributed randomly across the hybrid cell surface membrane.

The diagram represents the results of the experiment by showing the positions of the human and mouse proteins on the surface of the cells.



What does this experiment show?

A Movement of the phospholipids pushes the membrane proteins apart.

B Some membrane proteins move through the phospholipids to different places.

C The phospholipids of the human and mouse cell surface membranes do not mix.

D The proteins of human cell surface membranes can move further than those of mouse cells.

ANS: B

10 Batrachotoxin is a poison found in frogs in the Columbian jungle. The poison is used by Native Indians to produce poison darts. The poison works by increasing the permeability of some cell surface membranes to sodium ions, which move out of the cells. Which statements are correct for cells affected by batrachotoxin?

1 The intracellular fluid has a less negative water potential than the extracellular fluid.

2 The extracellular fluid has a less negative water potential than the intracellular fluid.

3 Water leaves the cells by osmosis, causing the cells to shrink.

4 Water enters the cells by osmosis, causing the cells to swell.

A **1** and **3**

B **1** and **4**

C **2** and **3**

D **2** and **4**

ANS: A

11 DNA replication in eukaryotes involves the following processes.

- RNA primer molecules are attached to each strand at points of origin of replication.
- DNA polymerase attaches to primers and synthesises new strands of DNA in a 5' to 3' direction.
- One strand, called the leading strand, is synthesised in continuous long sections.
- The other strand, called the lagging strand, is synthesised in short sections.
- RNA primers are replaced by DNA nucleotides on both strands.

Which statement explains the difference in the way in which the two strands of a DNA molecule are synthesised?

A DNA polymerase enzymes can only synthesise DNA in one direction.

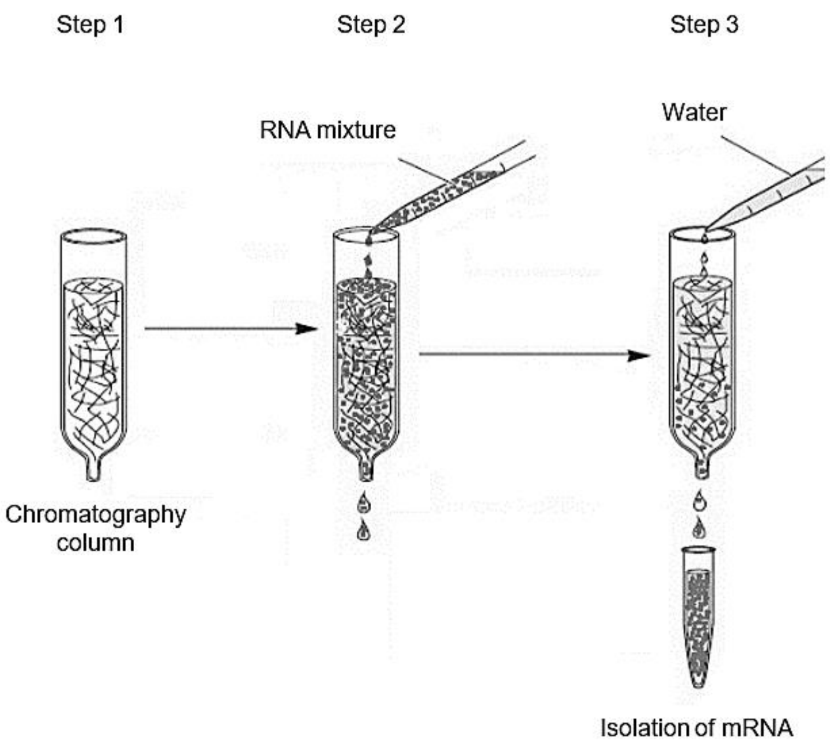
B Fewer RNA primers are needed on the leading strand.

C The lagging strand has more binding points for RNA primers.

D The replication of DNA is semi-conservative. .

ANS: A

12 The isolation of mRNA can be done by passing an RNA mixture obtained from homogenised tissue through a chromatography column. The procedure of the mRNA isolation is as follows:



Step 1: Set up a chromatography column which contains short lengths of uracil nucleotides attached to a solid support medium.

Step 2: Add RNA mixture to the chromatography column. RNA that do not hybridise with the uracil nucleotides will pass through and leave the column.

Step 3: Add water to the chromatography column to remove and isolate the hybridised mRNA.

Which of the following statements is **not** true?

A rRNA and tRNA molecules could pass through and leave the column in Step 2

B Complementary base pairing is responsible for the attachment of RNA to the solid support medium.

C The isolated mRNA in Step 3 do not contain introns.

D The RNA that attached to the chromatography column lack poly-(A) tails.

ANS: D

13 A polypeptide has the amino acid sequence glycine – arginine – lysine – serine.

The table below gives possible tRNA anticodons for each amino acid.

Amino Acid	Possible tRNA anticodons	
Arginine	3' UCC 5'	3' GCG 5'
Glycine	3' CCA 5'	3' CCU 5'
Lysine	3' UUC 5'	3' UUU 5'
Serine	3' AGG 5'	3' UCG 5'

Which sequence of bases on DNA would code for the polypeptide?

A 3' CCA CGC AAG AGC 5'

B 3' CCT TCC TTC TCG 5'

C 5' GGA AGG AAA AGC 3'

D 5' GGT TGG TTG TGC 3'

ANS: B

14 Which of the following shows the possible effects of a single nucleotide substitution in each of the following locations in a gene on the production of the protein it codes for?

	Promoter	Transcription terminator	Start codon	Stop codon	Middle of an intron
A	No protein product is produced	Protein product is shorter than normal	Protein product is longer than normal	Protein product is normal	Too much protein product is produced
B	Too much protein product is produced	Protein product is normal	No protein product is produced	Protein product is longer than normal	Protein product is normal
C	Protein product is normal	Protein product is longer than normal	Protein product is shorter than normal	Too much protein product is produce	Protein product is longer than normal
D	Protein product is longer than normal	Too much protein product is produced	Protein product is normal	Protein product is shorter than normal	No protein product is produced

15 Which of the following statements about the cell cycle is correct?

A G_1 is relatively the longest phase due to the duplication of organelles and checking of DNA damage.

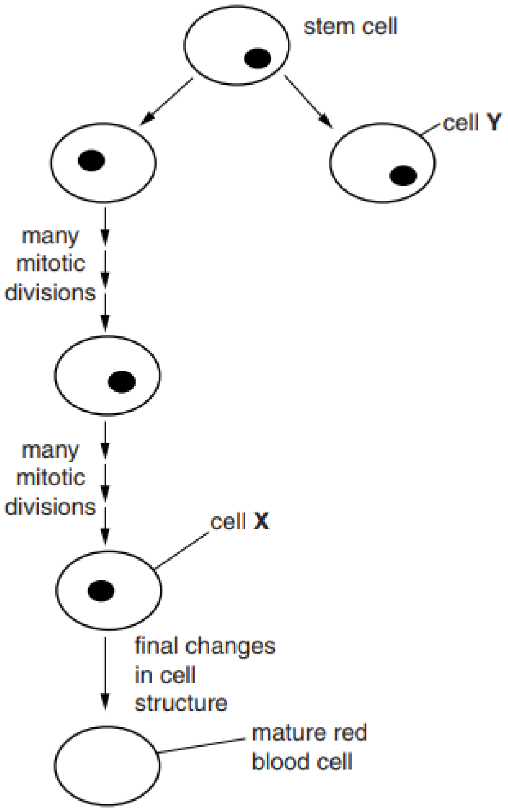
B G_2 phase occurs before Interphase of the cell cycle.

C M phase represents the phase where mitosis occurs.

D G_0 occurs between any phases within the cell cycle.

ANS: A

16 Bone marrow contains many stem cells. Some of these stem cells are responsible for the replacement of red blood cells. During the production of red blood cells, a series of changes occur to the cell structure. The figure below shows the production of a red blood cell from one of these stem cells.



Which of the following correctly describes the changes that occur as cell **X** becomes a mature biconcave red blood cell?

1 displays cell surface glycoproteins such as ABC antigens?

2 becomes multipotent

3 synthesises haemoglobin and carbonic anhydrase

4 loses its nucleus

5 loses organelles such as ribosomes, ER, mitochondria

6 loses ability to divide indefinitely

A **1, 2, 4, 6**

B **2, 3, 4, 6**

C **1, 3, 4, 5**

D **3, 4, 5, 6**

ANS: D

The following information is used for both questions **17** and **18**.

Development of gene editing tools opened up the possibility of correcting mutations associated with genetic disorders. For example, induced pluripotent stem cells (iPSCs) from patients with Alzheimer's disease can be formed from cells extracted from the patient and their DNA edited to correct the mutation. These cells are then re-introduced back to the same patient to treat the disease. Such an approach can accurately edit the DNA with an efficiency of up to 90% of the stem cells treated.

17 Which statement is true for iPSCs?

A iPSCs have the potential to become many adult cell types, including cells of the extra-embryonic membranes.

B iPSCs are harvested from the inner cell mass of the blastocyst.

C iPSCs can differentiate into various cell types by altering culture conditions or introducing specific transcription factors.

D iPSCs can undergo self-renewal to replace the first 16 cells produced by mitotic division of the zygote.

ANS: C

18 Which statement is **not** an ethical implication of the use of such gene editing tools?

A The efficacy and safety of the method are not well understood yet.

B High costs of gene editing may result in inequality between the rich and the poor.

C Gene editing in germline cells can lead to artificial enhancement of traits for subsequent generations.

D Further studies are required to understand the specific culture conditions for cell differentiation.

ANS: D

19 The germ cells of an organism contain six chromosomes (2n = 6), with an average of 18 units of DNA per chromatid.

The table below shows the results of measuring the amount of DNA in the germ cells of this organism at different stages of mitosis.

Which of the following shows the correct amount of DNA and number of chromosomes in the cell during a particular stage?

	Units of DNA per cell	Number of chromosomes per cell	Stage of nuclear division
A	36	3	Telophase
B	54	3	Metaphase
C	108	6	Prophase
D	216	12	Anaphase

ANS: D

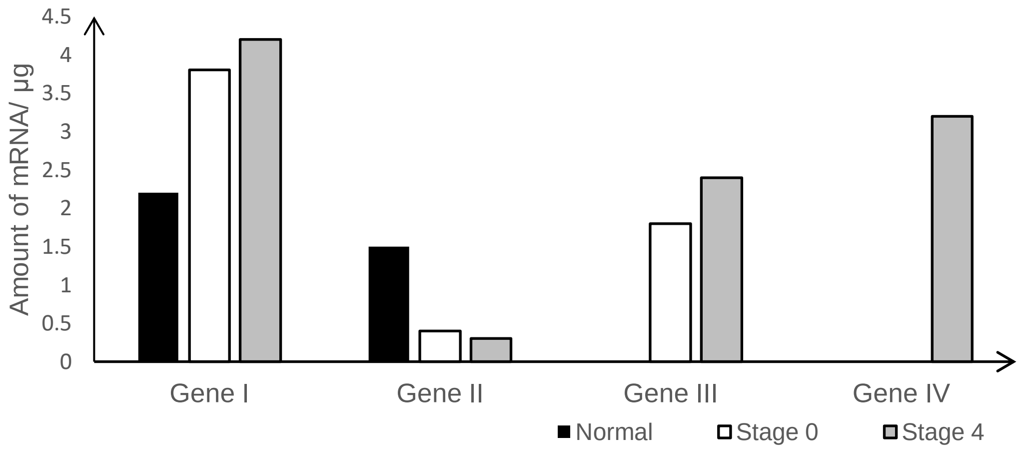
20 When a patient is diagnosed with cancer, the diagnosis will include the stage of the cancer. There are typically five stages.

Stage	Description
0	Small localised tumour mass which has not spread to nearby tissues.
1	Small localised tumour mass that has not grown deeply into nearby tissues. It has not spread to the lymph nodes or other parts of the body.
2 and 3	Larger tumour mass which has grown more deeply into nearby tissues. It may has also spread to lymph nodes but not to other parts of the body.
4	Cancer has spread to other organs or parts of the body.

The expression of genes **I – IV** was studied in three different individuals:

- Normal, not suffering from cancer
- Stage 0 cancer
- Stage 4 cancer

These genes have been implicated in cancer development. The amount of mRNA transcribed from these genes was quantified.

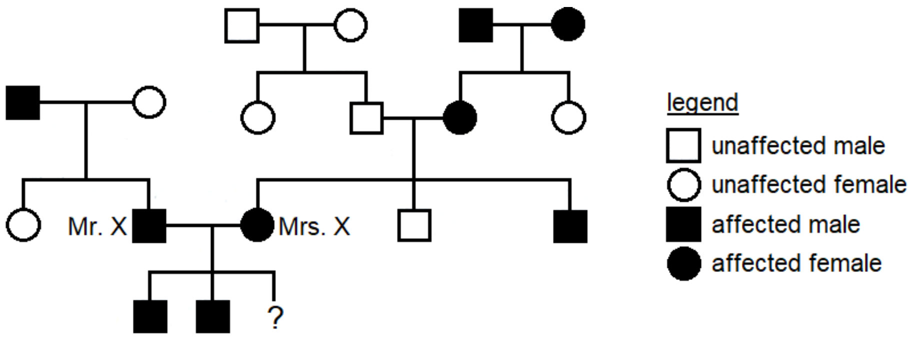


Which of the following is likely to be the correct identities of genes **I** to **IV**?

	Gene I	Gene II	Gene III	Gene IV
A	Proto-oncogene	Tumour suppressor gene	Gene coding for telomerase	Gene involved in angiogenesis
B	Proto-oncogene	Tumour suppressor gene	Gene involved in angiogenesis	Gene involved in contact inhibition
C	Tumour suppressor gene	Proto-oncogene	Gene coding for telomerase	Gene involved in angiogenesis
D	Tumour suppressor gene	Proto-oncogene	Gene involved in angiogenesis	Gene involved in anchorage dependence

ANS: A

21 The family tree below shows the inheritance of a heart disease due to hypercholesterolaemia.



Mrs. X is expecting a third child. If the child is a son, what is the probability that he will be unaffected?

A 0%

B 12.5%

C 25%

D 50%

ANS: C

22 In cattle, the gene responsible for normal development of hair and teeth, ectodysplasin 1 (ED1) gene, is located on the X chromosome. Mutations in the ED1 gene result in a rare genetic disorder called anhidrotic ectodermal dysplasia.

Another character, the presence of horns, is determined by a gene on an autosome. The allele for the absence of horns (H) is dominant and the allele for the presence of horns (h) is recessive.

A horned bull with anhidrotic ectodermal dysplasia was mated on several occasions to the same female. A large number of offspring consisting of males and females in equal numbers in all combinations of phenotypes are shown in the table.

Offspring phenotypes
No anhidrotic ectodermal dysplasia, horns present
No anhidrotic ectodermal dysplasia, horns absent
Anhidrotic ectodermal dysplasia, horns present
Anhidrotic ectodermal dysplasia, horns absent

If $\mathbf{X^E}$ represents an X chromosome carrying the normal *ED1* allele and $\mathbf{X^{\underline{e}}}$ represents an X chromosome carrying the *ED1* allele for anhidrotic ectodermal dysplasia, what is the genotype of the female parent?

A X^EX^EHH

B X^EX^EHh

C $X^EX^{\underline{e}}HH$

D $X^EX^{\underline{e}}Hh$

ANS: D

23 In an experiment to study photosynthesis, leaf cells were homogenised to break their cell walls.

The resulting leaf cell suspensions containing the cytoplasm and organelles were then placed in test-tubes containing non-labeled water ($H_2{}^{16}O$) and ^{18}O-labeled water ($H_2{}^{18}O$) respectively.

DCPIP is a hydrogen acceptor and will turn from blue to colourless when it is reduced. A few drops of DCPIP were added to each test-tube above.

Which of the following shows the results of the two test-tubes after they were exposed to light for 30 minutes?

	test-tube with $H_2{}^{16}O$		test-tube with $H_2{}^{18}O$	
	gas evolved	colour observed	gas evolved	colour observed
A	$C^{16}O_2$	blue	$C^{18}O_2$	blue
B	$^{16}O_2$	blue	$^{16}O_2$	blue
C	$C^{16}O_2$	green	$C^{18}O_2$	green
D	$^{16}O_2$	green	$^{18}O_2$	green

ANS: D

24 Which of these statements about aerobic respiration is correct?

A NADH and FADH are needed in the processes of respiration within and outside of the mitochondria.

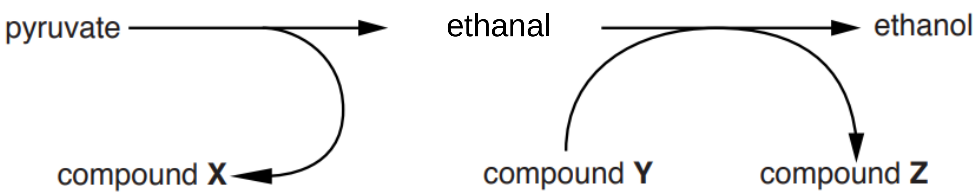
B The conversion of glycerate 3-phosphate to glyceraldehyde 3-phosphate is part of the process that yields pyruvate at the end of glycolysis.

C At the end of glycolysis, pyruvate is converted to acetyl CoA and transported into the mitochondrial matrix.

D ATP is generated by substrate level phosphorylation resulting in the formation of succinate.

ANS: D

25 The flowchart outlines the process of anaerobic respiration in yeast cells.



Which of the statements regarding the identities of the compounds are correct?

1 Compound **X** is a waste product.

2 Compound **Y** is necessary for more rounds of glycolysis.

3 Compound **Z** acts as an oxidising agent.

A **1, 2** and **3**

B **1** and **2** only

C **1** and **3** only

D **2** and **3** only

ANS: C

26 Researchers have found evidence of natural selection in humans.

1 In ancestral populations, only babies and children needed to digest the milk sugar, lactose. The gene coding for the enzyme lactase (*LCT* gene) was switched off before adulthood.

2 Today, lactose-intolerant adults cannot digest lactose, leading to abdominal pain after eating food containing lactose.

3 A mutation in the *LCT* gene results in adults to be able to digest lactose. This is called lactose persistence.

4 Lactose persistence increased in populations in Europe several thousand years ago.

5 The increase in lactose persistence in Europe coincided with an increase in farming of cows for milk.

Which of the following statements does **not** explain why there was selection for lactose persistence in humans several thousand years ago?

A Reliance on milk products selected for individuals who were able to digest lactose due to the presence of the mutated *LCT* gene in them.

B Compared to lactose-intolerant individuals, individuals with lactose persistence are likely to get fewer side effects after consuming food containing lactose.

C On average, more individuals who produce high levels of lactase in their lifetime live to reproductive age, as they were able to utilise an additional source of nutrient.

D Individuals with lactose persistence consumed more milk than the rest of the population and passed on the mutant allele to their offspring.

ANS: D

27 Deer mice are small, ground-dwelling rodents. They normally have dark coats.

Some paler coated mice have been observed in an area where new sand hills were formed between 8000 and 15 000 years ago. The sand hills are sparsely covered with scrubby plants.

Studies have shown that the change in coat colour was due to a mutation in a gene, causing a single amino acid deletion from the protein for the coat colour pigment. This mutation seems to be spreading through the population.

Which statement most fully explains how the evolution of this species at this site has occurred?

A Better camouflage increases the chance of survival by reducing predation.

B Deer mice which live longer usually leave more offspring since they have more reproductive opportunities.

C The occurrence of the mutation provided new variation on which natural selection can act.

D The paler colour gives better camouflage against the background of the new sand hills.

ANS: C

28 Approximately 1 in 20 Europeans are heterozygous for a recessive allele responsible for the genetic condition, cystic fibrosis (CF). People who are homozygous for CF have a reduced life expectancy.

Heterozygotes are more resistant to some bacterial infections of the gut, such as typhoid fever, than homozygotes for the normal, dominant allele.

What could explain for the high incidence of the recessive CF allele in the European population?

A natural selection favouring heterozygotes

B natural selection favouring homozygotes for the recessive CF allele

C lack of genetic drift in the European population

D mutation

ANS: A

29 Which statement concerning chrysanthemum plants, of the genus *Dendranthema*, is a valid example of how the environment may affect the phenotype?

A Anthocyanins and anthoxanthins are vacuolar pigments, whereas xanthophylls and carotenes are pigments found in membrane-bound organelles known as plastids. These, together with molecules known as co-pigments, are responsible for the variation observed in petal colour in *Dendranthema*.

B Identical genetic crosses performed between varieties of *Dendeanthema* result in a greater proportion of offspring plants with plastids exhibiting a yellow colour when grown in a field and a greater proportion of offspring plants with colourless plastids when grown in a glasshouse.

C The seeds of a cross between *Dendranthema weyrichii* and *Dendranthema grandiflora* produce plants that are far more frost-tolerant and exhibit an extended flowering season compared with both parent plants.

D The seeds of a cross between *Dendranthema weyrichii* (height varying between 12.5 – 15.0 cm) and *Dendranthema grandiflora* (height varying between 8.0 – 25.0 cm) produce plants, when grown in natural day length, of a height varying between 55.0 – 71.0 cm.

ANS: B

30 Which of the following **does not** explain why the population is the smallest unit that can evolve?

A Natural selection involves competition between individuals in a population.

B Evolution occurs when allele frequency in a population changes due to selection or chance events like infections with lethal diseases.

C Differential reproductive success is observed at the population level due to the phenotypic variations in the population.

D Evolution involves the introduction of advantageous mutations into the gene pool of a population as a result of a selective pressure.

ANS: D

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